Matrices and notation

Matrix: $A = [a_{ij}]$, a rectangular array of mn real numbers in m horizontal rows and n vertical columns; size $m \times n$ ("m by n").

Entry: a_{ij} is in row i, column j; $1 \le i \le m$, $1 \le j \le n$.

Row i: $\begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{bmatrix}$.

Column j: $\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$.

Equality: $A = B$ if and only if they have the same size and $a_{ij} = b_{ij}$ for every entry.

Entrywise arithmetic and its rules

Let $A = [a_{ij}]$, $B = [b_{ij}]$, and C be $m \times n$ matrices; $r, s \in \mathbb{R}$.

Sum: $A + B = [a_{ij} + b_{ij}]$; defined only for matrices of the same size.

Scalar multiple: $rA = [ra_{ij}]$; size remains $m \times n$.

Zero matrix: O, the $m \times n$ matrix of all zeros; $A + O = A$.

Negative: $-A = (-1)A$; $A + (-A) = O$.

Commutativity of addition: $A + B = B + A$.

Associativity of addition: $A + (B + C) = (A + B) + C$.

Scalar multiplication rules:

$$r(sA) = (rs)A, \qquad (r + s)A = rA + sA, \qquad r(A + B) = rA + rB.$$

Transpose and special matrices

Transpose: $A^T = [a_{ij}^T]$, with $a_{ij}^T = a_{ji}$; interchanges rows and columns. If A is $m \times n$, then A^T is $n \times m$.

Vectors under transpose: column vectors become row vectors and vice versa.

Transpose rules: for A, B of the same size and $r \in \mathbb{R}$,

$$(A^T)^T = A, \qquad (A + B)^T = A^T + B^T, \qquad (rA)^T = rA^T.$$

Main diagonal: entries $a_{11}, a_{22}, \ldots, a_{kk}$, where $k = \min(m, n)$.

Diagonal matrix: all entries off the main diagonal are zero.

Square matrix: equal numbers of rows and columns ($m = n$).

Symmetric matrix: a real square matrix A with $A^T = A$.